

UNIT I — Lecture Module 3

CLOUD HARDWARE AND SOFTWARE**INTRODUCTION:**

Cloud computing is not an abstract concept — it is built on real, tangible physical infrastructure combined with sophisticated software layers that abstract and manage that infrastructure. Understanding the hardware and software foundations of cloud computing is essential for designing, deploying, and optimising cloud solutions.

The hardware layer consists of the physical building blocks: servers, storage devices, networking equipment, power systems, and cooling infrastructure housed in massive datacenters. The software layer sits above the hardware and provides the intelligence: virtualisation, resource scheduling, identity management, storage abstraction, networking virtualisation, monitoring, and self-service interfaces. Together, these hardware and software components form the cloud computing infrastructure stack — a set of hierarchical layers where each level abstracts the complexity of the layer below, ultimately presenting the consumer with simple, programmable, on-demand resources.

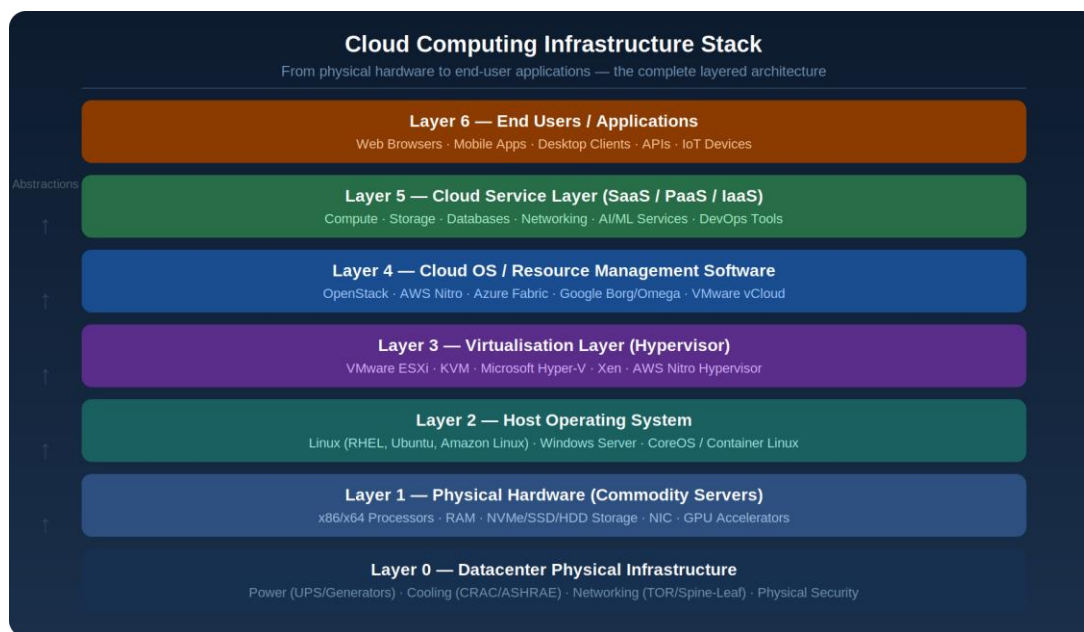


Fig : Cloud Computing Infrastructure Stack — from physical datacenter to end-user applications

Figure 1 illustrates the complete six-layer cloud infrastructure stack. The bottom layers (0 and 1) represent physical infrastructure. Layer 2 provides the host operating system. Layer 3 (the hypervisor) creates the virtualisation boundary. Layer 4 is the cloud management software. Layer 5 delivers cloud services (IaaS/PaaS/SaaS). Layer 6 represents end users and their applications. Each upper layer depends entirely on the correctness and availability of all layers beneath it.

Cloud Datacenter — Physical Infrastructure:

A cloud datacenter is a specialised facility purpose-built to house, power, cool, connect, and secure thousands or tens of thousands of servers and networking devices. It is fundamentally different from a traditional corporate server room in scale, design philosophy, redundancy, and operational efficiency.

Hyperscale datacenters — those operated by major cloud providers such as Amazon Web Services, Microsoft Azure, and Google Cloud — are among the largest, most complex engineered structures in existence. A single hyperscale campus may cover 200,000+ square metres, consume 100 megawatts or more of electrical power, and house hundreds of thousands of physical servers operating continuously.

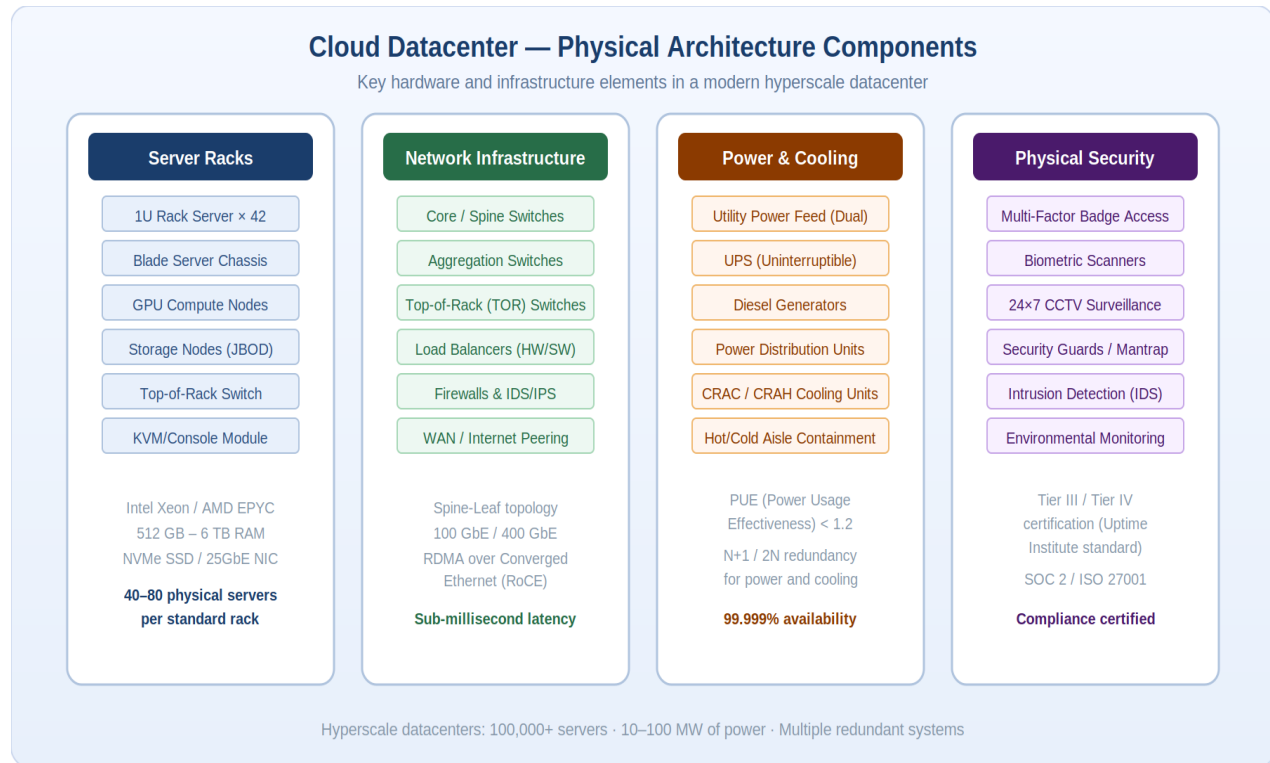


Fig : Cloud Datacenter Physical Architecture — Server Racks, Networking, Power & Cooling, and Security

SERVER HARDWARE:

Cloud datacenters use commodity x86/x64 servers — standardised, off-the-shelf computing nodes built from high-volume, cost-effective components rather than proprietary, vendor-specific hardware. This design choice enables massive economies of scale and simplifies replacement.

- Processor (CPU):** Modern cloud servers use Intel Xeon Scalable or AMD EPYC processors. These server-grade CPUs support virtualisation extensions (Intel VT-x, AMD-V), which are required for running hypervisors. A single 2-socket server may contain 128–256 physical CPU cores.
- Memory (RAM):** Cloud servers use DDR5 ECC (Error-Correcting Code) RAM. ECC detects and corrects single-bit memory errors automatically, ensuring reliability for long-running virtualised workloads. A single server node may carry 512 GB to 6 TB of RAM.

- **Storage:** Three types of storage are common: NVMe SSDs (Non-Volatile Memory Express — ultra-low latency, high IOPS), SATA SSDs (mid-tier cost/performance), and HDDs (high-capacity archival storage). NVMe drives are connected directly to the CPU via PCIe lanes.
- **Network Interface Card (NIC):** 25 GbE or 100 GbE NICs connect servers to the datacenter network fabric. Some servers use RDMA-capable (Remote Direct Memory Access) NICs for ultra-low latency distributed computing.
- **GPU Accelerators:** For AI/ML, scientific computing, and graphics workloads, servers may include NVIDIA A100, H100, or AMD Instinct GPUs. A single GPU node may contain $8 \times$ A100 GPUs with 640 GB of GPU memory.

DATACENTER NETWORK ARCHITECTURE:

The network within a cloud datacenter must provide extremely high bandwidth, very low latency, and full redundancy. Modern cloud datacenters use a Spine-Leaf (Clos) network topology rather than the traditional hierarchical three-tier model.

Network Layer	Component	Speed	Role
Core / Spine	Core Switches	400 GbE	High-speed backbone interconnecting all leaf switches
Aggregation / Leaf	ToR Switches	25–100 GbE	Connect server racks to the spine; one switch per rack
Server Access	NIC / HBA	25–100 GbE	Physical link from server to top-of-rack switch
WAN / Internet	Border Routers / BGP	10–100 Gbps	Connect datacenter to Internet exchange points

Load Balancing	Hardware / Variable Software LBs	Distribute incoming traffic across multiple servers
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POWER INFRASTRUCTURE:

Continuous, reliable power delivery is critical for cloud datacenters. A power interruption lasting even a few milliseconds can corrupt data and disrupt millions of customers. Cloud datacenters employ multiple layers of power redundancy:

- **Utility power feeds:** Datacenters are connected to the electrical grid via two or more independent utility feeds from separate substations, providing N+1 or 2N power redundancy.
- **UPS (Uninterruptible Power Supply):** Large battery banks (or flywheels) provide instantaneous power backup during the transition from utility to generator power — typically 10–30 seconds of runtime.
- **Diesel generators:** Massive diesel generators (each capable of powering thousands of servers) automatically start within 10–30 seconds of a utility failure and can run for days on stored fuel.
- **PDU (Power Distribution Units):** Rackmount PDUs distribute power from the facility electrical panels to individual servers, with monitoring per-outlet for granular power measurement.
- **PUE (Power Usage Effectiveness):** $PUE = \text{Total Facility Power} \div \text{IT Equipment Power}$. A PUE of 1.0 means 100% of power goes to computing. Google's global average PUE is 1.10; industry average is ~1.58.

COOLING INFRASTRUCTURE:

Servers generate enormous heat — a single rack of high-density compute nodes can produce 20–50 kW. Effective cooling is critical to server reliability and energy efficiency.

- **Hot/Cold Aisle Containment:** Server racks are arranged in alternating rows facing front-to-front (cold aisles) and back-to-back (hot aisles). Cold air is supplied to the cold aisles; hot exhaust air is captured from the hot aisles and returned to cooling units.
- **CRAC/CRAH Units:** Computer Room Air Conditioning (CRAC) and Computer Room Air Handlers (CRAH) are precision cooling units that maintain temperature and humidity within strict tolerances (18–27°C, 40–60% RH per ASHRAE A1 class).
- **Liquid Cooling:** For ultra-high-density GPU clusters (20–100 kW per rack), direct liquid cooling (DLC) runs chilled water or dielectric fluid through cold plates attached to CPUs and GPUs. More energy-efficient than air cooling.
- **Free Cooling:** In cold climates, datacenters use outside air (economisers) to cool the facility without mechanical refrigeration for a significant portion of the year, dramatically reducing cooling energy costs.

Virtualisation — The Core Enabling Technology:

Virtualisation is the foundational technology that makes cloud computing economically and practically feasible. Without virtualisation, cloud computing as we know it — with its multi-tenancy, elasticity, and pay-per-use model — would not be possible. Virtualisation allows a single physical machine to run multiple independent virtual machines (VMs) simultaneously, each with its own isolated operating system, applications, storage, and network interfaces.

3.1 What is Virtualisation?

Virtualisation is the process of creating a software-based (virtual) representation of hardware resources. A layer of software called a hypervisor (or Virtual Machine Monitor — VMM) intercepts all hardware access requests from guest operating systems and maps them to the underlying physical hardware. This abstraction creates the illusion for each guest OS that it has exclusive access to dedicated hardware, while in reality multiple guests share the same physical resources.

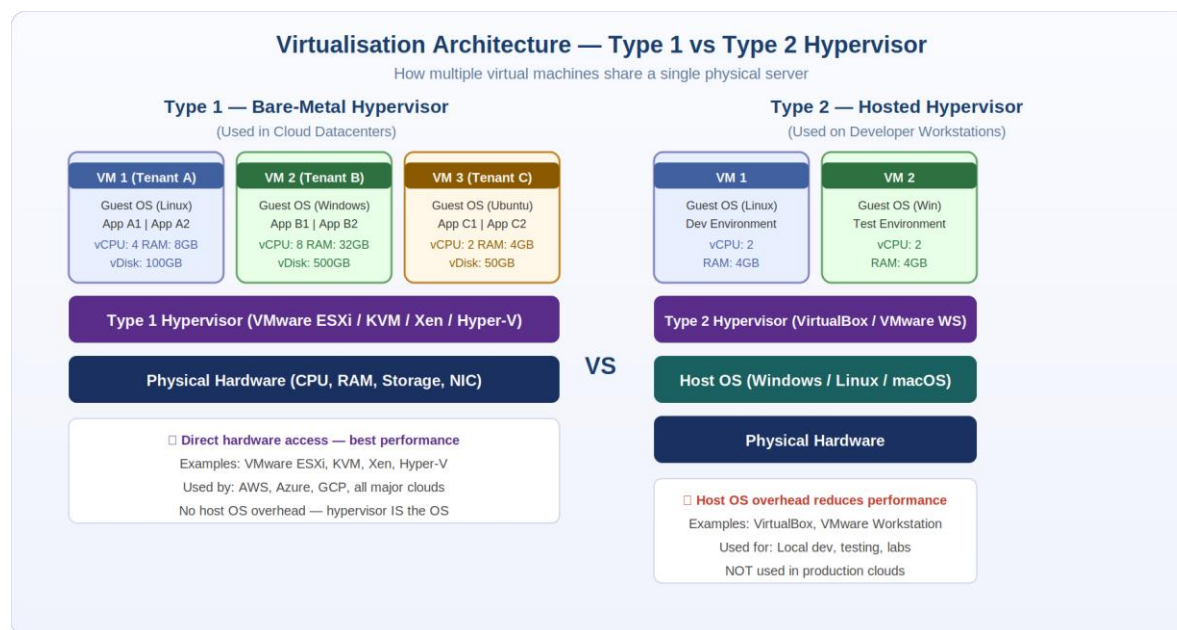


Figure : Virtualisation Architecture — Type 1 (Bare-Metal) vs Type 2 (Hosted) Hypervisor

TYPE 1 VS TYPE 2 HYPERVISORS:

Hypervisors are classified into two types based on whether they run directly on hardware or on top of a host operating system.

Dimension	Type 1 — Bare-Metal Hypervisor	Type 2 — Hosted Hypervisor
Installation	Installed directly on physical hardware; no host OS	Installed on top of a host OS (Windows/Linux/macOS)
Performance	Near-native; direct hardware access	Lower; host OS overhead reduces throughput
Use case	Production cloud datacenters; enterprise servers	Developer workstations; labs; testing
Resource efficiency	Very high; no host OS wasting RAM/CPU	Lower; host OS consumes significant resources
Boot time	Seconds (hypervisor boots like an OS)	Slower; host OS must boot first, then hypervisor
Fault isolation	Host OS crash cannot affect VMs	Host OS crash terminates all VMs

Examples	VMware ESXi, KVM, Xen, Microsoft Hyper-V Server	Oracle VirtualBox, VMware Workstation, Parallels
Used by clouds	Yes — all major cloud providers use Type 1	No — not used in production cloud datacenters

CLOUD MANAGEMENT SOFTWARE:

While physical hardware provides the raw compute, storage, and networking capacity, it is the cloud management software that transforms this hardware into a usable, automated, multi-tenant cloud service. The cloud management software is responsible for resource orchestration, tenant isolation, automated provisioning, billing, monitoring, and self-service access.

This collection of software components is collectively referred to as the Cloud Management Platform (CMP). Different cloud providers have developed their own CMPs (AWS Nitro + EC2 control plane, Azure Fabric, Google Borg/Omega), while open-source projects like OpenStack provide a vendor-neutral alternative

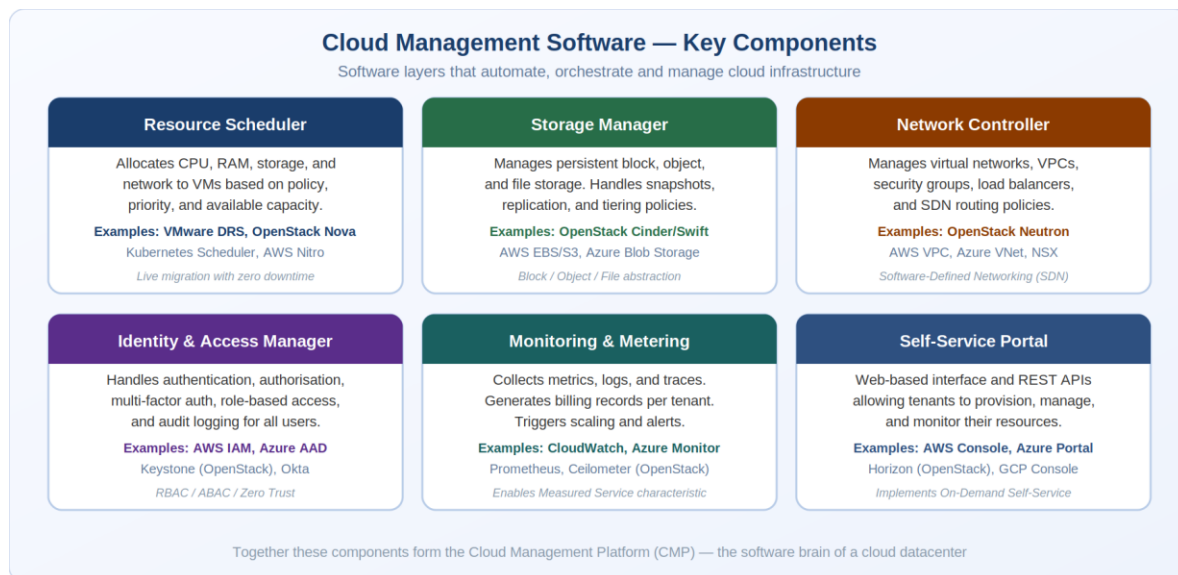


Figure : Cloud Management Software — six core components of the Cloud Management Platform (CMP)

STORAGE MANAGER:

Cloud storage is far more complex than simply attaching hard drives to servers. The storage manager abstracts physical storage devices into logical storage services that can be provisioned, snapshotted, replicated, and tiered automatically.

Storage Type	Description	Cloud Example
Block Storage	Raw storage volumes attached to VMs; formatted with a filesystem by the guest OS	AWS EBS, Azure Managed Disks, Cinder (OpenStack)
Object Storage	Flat namespace of immutable objects (files/blobs) accessed via HTTP REST API	AWS S3, Azure Blob Storage, Swift (OpenStack)
File Storage	Shared network filesystem (NFS/SMB) mountable by multiple VMs simultaneously	AWS EFS, Azure Files, Manila (OpenStack)
Ephemeral	Temporary local storage on the VM's physical host — lost when VM is terminated	AWS Instance Store, local disk

NETWORK CONTROLLER (SOFTWARE-DEFINED NETWORKING):

Traditional networks are configured manually on physical switches and routers using vendor-specific CLI commands. Cloud networking requires a fundamentally different approach — Software-Defined Networking (SDN) — which separates the network control plane (where routing decisions are made) from the data plane (where packets are forwarded), allowing the network to be programmatically controlled and reconfigured in real time.

- **Virtual Private Cloud (VPC):** Each tenant receives a logically isolated virtual network segment with its own IP address range, routing tables, security groups, and internet gateway. VPCs are the fundamental network isolation boundary in cloud computing.
- **Security Groups:** Virtual stateful firewalls that control inbound and outbound traffic at the VM level based on protocol, port, and source/destination IP rules.
- **Virtual Load Balancers:** Software-based load balancers distribute incoming requests across a pool of backend VM instances. Unlike physical load balancers, they scale automatically and are provisioned instantly.

- **Overlay Networks:** VMs communicate over a virtual network layer (VXLAN, Geneve) overlaid on top of the physical network, enabling tenant network isolation regardless of the physical infrastructure topology.

IDENTITY AND ACCESS MANAGEMENT (IAM):

In a multi-tenant cloud environment, controlling who can access what resources is critical for both security and compliance. IAM provides the authentication (verifying who you are) and authorisation (determining what you are allowed to do) framework for all cloud operations.

- **Authentication:** Users prove their identity via username/password, API keys, X.509 certificates, or federated identity (SAML, OAuth 2.0, OpenID Connect) with external identity providers (Active Directory, Google Workspace).
- **Multi-Factor Authentication (MFA):** Requires a second factor (TOTP app, SMS OTP, hardware key) in addition to a password, protecting against credential theft.
- **Role-Based Access Control (RBAC):** Users are assigned roles (e.g., Administrator, Developer, ReadOnly) that define which API operations and resources they can access.
- **Attribute-Based Access Control (ABAC):** Fine-grained policies based on attributes of the user, resource, and environment (e.g., 'allow access only during business hours from corporate IP range').
- **Service Accounts:** Non-human identities (IAM roles) assigned to applications and services, allowing VMs and serverless functions to access other cloud services without embedding credentials in code.

COMMODITY HARDWARE STRATEGY IN CLOUD COMPUTING:

One of the most important strategic decisions in cloud computing infrastructure is the deliberate use of commodity (standard, off-the-shelf) hardware rather than proprietary, vendor-specific equipment. This philosophy, pioneered by companies like Google and later adopted by all major cloud providers, has profound implications for cost, reliability, scalability, and innovation.

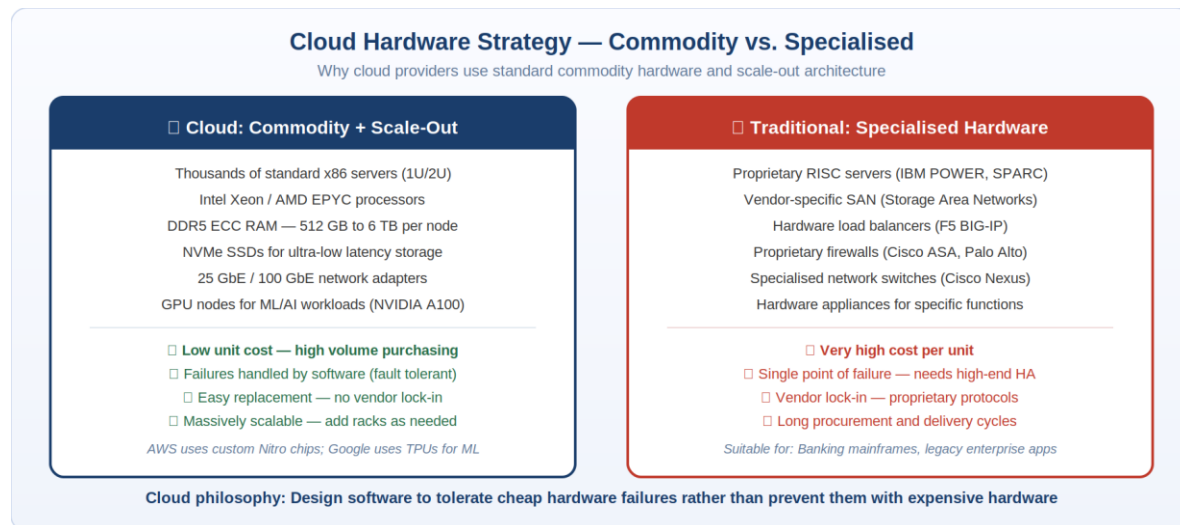


Fig: Cloud Commodity Hardware vs. Traditional Specialised Hardware — strategic comparison

PRINCIPLES OF COMMODITY HARDWARE DESIGN:

- **Design for failure:** Instead of spending enormous amounts on ultra-reliable hardware (which still fails eventually), cloud providers use standard hardware and design software systems to automatically detect and recover from hardware failures. If a disk fails, the storage system reconstructs the data from replicas. If a server fails, the scheduler migrates its VMs to healthy hosts.
- **Scale horizontally:** Rather than buying one very large, very expensive server (scale-up), cloud providers buy thousands of smaller commodity servers (scale-out). Horizontal scaling is more cost-effective and provides natural fault tolerance.
- **High-volume purchasing:** By purchasing hardware at enormous scale (millions of units), cloud providers negotiate prices far below what any individual enterprise can achieve, further reinforcing the cost advantage.
- **Custom silicon:** At sufficient scale, cloud providers develop custom chips optimised for their specific workloads — AWS Graviton (ARM-based CPU), AWS Nitro (dedicated security/IO processor), Google TPU (Tensor Processing Unit for ML), Microsoft Azure Maia (AI accelerator). This further reduces costs and improves performance beyond what commodity CPUs offer.

HARDWARE FAILURES AND RELIABILITY STRATEGY:

In a datacenter with 100,000 servers, statistical failure rates mean that multiple hardware components fail every day. Cloud providers embrace this reality and build reliability into their software rather than their hardware.

Failure Type	Traditional Response	Cloud Response
Single disk failure	Expensive RAID hardware; manual rebuild	Software replication (3× copies); automatic transparent recovery
Server hardware fault	Emergency vendor support; RMA process	VM live-migrated off failing host; failed server queued for replacement
Network switch failure	Redundant switch pair (active/standby)	Spine-leaf topology; multiple equal-cost paths; automatic failover via ECMP
Power failure	UPS + generator + redundant feeds	Same + geographic region redundancy; AZ isolation; cross-region DR

Datacenter disaster	Off-site backup tapes	Multi-region deployment; replication	active-active continuous
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OPENSTACK — OPEN-SOURCE CLOUD MANAGEMENT PLATFORM:

OpenStack is the leading open-source cloud computing platform, providing a complete software stack for building and managing public and private cloud infrastructure. Originally launched in 2010 as a collaboration between Rackspace and NASA, OpenStack is now maintained by the OpenStack Foundation with contributions from hundreds of organisations including Red Hat, IBM, Intel, and Ericsson.

OPENSTACK ARCHITECTURE:

OpenStack components communicate with each other through REST APIs and a message bus (typically RabbitMQ). This loosely coupled, API-first architecture allows each component to be independently scaled, upgraded, or replaced. The major architectural elements are:

- **Control Plane:** Runs the management services — Nova API, Keystone, Glance, Neutron Server, Heat API, Ceilometer. These are typically deployed on dedicated controller nodes with high availability (3+ nodes in a cluster using Pacemaker/Corosync).
- **Compute Nodes:** Physical servers running the KVM hypervisor and the nova-compute agent. These are the worker nodes that actually execute VMs. Scaled horizontally as capacity grows.
- **Network Nodes:** Run Neutron agents (L3 Agent, DHCP Agent) that implement routing, NAT, and DHCP for tenant virtual networks.
- **Storage Nodes:** Run Cinder (block storage) and/or Swift (object storage) services. Storage nodes may be separate from compute nodes or collocated depending on deployment model.
- **Message Bus:** RabbitMQ or ZeroMQ provides asynchronous communication between services, ensuring that temporary failures in one component do not cascade to others.

OPENSTACK VS. PROPRIETARY CLOUD:

OpenStack is functionally equivalent to the management platforms used by public cloud providers, but it is deployed and operated by the organisation itself rather than consumed as a service. This gives organisations full control over their infrastructure, data sovereignty, and customisation — at the cost of significant operational complexity. OpenStack is commonly used for private cloud deployments in telecommunications, government, and research institutions.

CLOUD STORAGE ARCHITECTURE:

Storage in cloud computing is designed from the ground up differently from traditional storage systems. Traditional enterprise storage relies on centralised, specialised storage arrays (SAN — Storage Area Networks, NAS — Network Attached Storage) that are expensive, proprietary, and difficult to scale. Cloud storage uses a distributed, software-defined approach that runs on commodity hardware and scales horizontally to exabytes.

DISTRIBUTED STORAGE PRINCIPLES:

- **Data replication:** Cloud storage systems store multiple copies (typically 3×) of every data object across different physical nodes, racks, or even datacenters. If one copy is lost (due to disk or node failure), the system automatically creates a new replica from one of the surviving copies, maintaining the replication factor.
- **Erasur coding:** A more storage-efficient alternative to full replication. Data is split into fragments and encoded with mathematical redundancy (e.g., Reed-Solomon coding), allowing the original data to be reconstructed even if several fragments are lost. AWS S3 uses erasure coding for durability.
- **Consistent hashing:** Used to distribute data objects across storage nodes in a balanced way, and to minimize data movement when nodes are added or removed from the cluster.
- **Eventual consistency vs. strong consistency:** Object storage systems (like S3) historically used eventual consistency — after a write, different readers might temporarily see different versions of the data. AWS S3 moved to strong read-after-write consistency in 2020.

CONTENT DELIVERY NETWORKS (CDN):

A CDN is a globally distributed network of cache servers (edge nodes) that stores copies of frequently accessed content (images, videos, static web assets) geographically close to end users. When a user requests a file, the CDN serves it from the nearest edge node rather than the origin server, dramatically reducing latency and origin server load.

- CDNs are a key component of the cloud software stack for any consumer-facing application.
- AWS CloudFront has 400+ edge locations worldwide; Akamai has 4,000+ points of presence.
- CDNs implement the Broad Network Access characteristic — making content available rapidly from anywhere.

SUMMARY — CLOUD HARDWARE AND SOFTWARE STACK :

This lecture module has covered the complete hardware and software foundations of cloud computing infrastructure. The table below consolidates the key topics across all sections for quick revision.

Layer	Component	Key Technologies	Purpose
Physical (L0)	Datacenter Facility	UPS, Generators, CRAC, Hot/Cold Aisle	Power, cool, and protect physical hardware
Physical (L1)	Server Hardware	Intel Xeon/AMD EPYC, NVMe, 100GbE NIC	Raw compute, storage, and networking capacity
Physical (L1)	Network Infrastructure	Spine-Leaf, VXLAN, SDN, Border Routers	High-bandwidth, low-latency interconnect
Virtualisation (L2/L3)	Hypervisor	KVM, VMware ESXi, Xen, Hyper-V, Nitro	Create and isolate virtual machines
Cloud OS (L4)	Resource Scheduler	OpenStack Nova, VMware DRS, K8s Scheduler	Optimal placement and live migration of VMs
Cloud OS (L4)	Storage Manager	Cinder (Block), Swift (Object), Glance	Abstract and provision storage services
Cloud OS (L4)	Network Controller	Neutron (OpenStack), AWS VPC, NSX	Virtual network isolation (SDN)
Cloud OS (L4)	IAM	Keystone, AWS IAM, Azure AAD	Authentication, authorisation, RBAC
Cloud OS (L4)	Monitoring & Metering	Ceilometer, CloudWatch, Prometheus	Usage tracking, billing, auto-scaling trigger
Cloud OS (L4)	Self-Service Portal	Horizon (OpenStack), AWS Console	On-demand provisioning interface
Service (L5)	IaaS / PaaS / SaaS	EC2, S3, RDS, Lambda, Office 365	Cloud services consumed by users